INCH-POUND
MIL-W-16878/27A(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/27(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 260°C, 1000 VOLTS, EXTRUDED INSULATION

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.

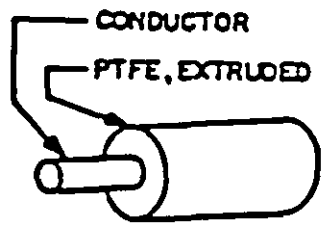


FIGURE 1. Construction.

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TABLE I. Construction details.

PIN 3/	Wire size	Strand-ing	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 1/ 2/	Coating		Min	Max
M16878/27-BAA*	32	1 X 32	Copper	Nickel	0.0080	0.034	0.042
M16878/27-DAA*	32	1 X 32	H.S.C.A.	Nickel	.0080	.034	.042
M16878/27-CAA*	32	1 X 32	C.C. steel	Nickel	.0080	.034	.042
M16878/27-BAB*	32	7 X 40	Copper	Nickel	.010	.036	.044
M16878/27-DAB*	32	7 X 40	H.S.C.A.	Nickel	.010	.036	.044
M16878/27-BBA*	30	1 X 30	Copper	Nickel	.0100	.036	.044
M16878/27-DBA*	30	1 X 30	H.S.C.A.	Nickel	.0100	.036	.044
M16878/27-CBA*	30	1 X 30	C.C. steel	Nickel	.0100	.036	.044
M16878/27-BBB*	30	7 X 38	Copper	Nickel	.012	.038	.046
M16878/27-DBB*	30	7 X 38	H.S.C.A.	Nickel	.012	.038	.046
M16878/27-BCA*	28	1 X 28	Copper	Nickel	.0126	.039	.047
M16878/27-DCA*	28	1 X 28	H.S.C.A.	Nickel	.0126	.039	.047
M16878/27-CCA*	28	1 X 28	C.C. steel	Nickel	.0126	.039	.047
M16878/27-BDA*	26	1 X 26	Copper	Nickel	.0159	.042	.050
M16878/27-DDA*	26	1 X 26	H.S.C.A.	Nickel	.0159	.042	.050
M16878/27-CDA*	26	1 X 26	C.C. steel	Nickel	.0159	.042	.050
M16878/27-BDB*	26	7 X 34	Copper	Nickel	.019	.045	.053
M16878/27-DDB*	26	7 X 34	H.S.C.A.	Nickel	.019	.045	.053
M16878/27-BEA*	24	1 X 24	Copper	Nickel	.0201	.046	.054
M16878/27-DEA*	24	1 X 24	H.S.C.A.	Nickel	.0201	.046	.054
M16878/27-CEA*	1/ 24	1 X 24	C.C. steel	Nickel	.0201	.046	.054
M16878/27-BEB*	24	7 X 32	Copper	Nickel	.024	.050	.058
M16878/27-DEB*	24	7 X 32	H.S.C.A.	Nickel	.024	.050	.058
M16878/27-BFA*	22	1 X 22	Copper	Nickel	.0254	.051	.060
M16878/27-DFA*	22	1 X 22	H.S.C.A.	Nickel	.0254	.051	.060
M16878/27-CFA*	1/ 22	1 X 22	C.C. steel	Nickel	.0254	.051	.060
M16878/27-BFB*	22	7 X 30	Copper	Nickel	.030	.056	.064
M16878/27-DFB*	22	7 X 30	H.S.C.A.	Nickel	.030	.056	.064
M16878/27-BGA*	20	1 X 20	Copper	Nickel	.032	.058	.066
M16878/27-DGA*	20	1 X 20	H.S.C.A.	Nickel	.032	.058	.066
M16878/27-BGB*	20	7 X 28	Copper	Nickel	.038	.064	.072
M16878/27-DGB*	20	7 X 28	H.S.C.A.	Nickel	.038	.064	.072
M16878/27-BHA*	18	1 X 18	Copper	Nickel	.0403	.066	.076
M16878/27-BHB*	18	7 X 26	Copper	Nickel	.048	.074	.084
M16878/27-BJA*	16	1 X 16	Copper	Nickel	.0508	.077	.089
M16878/27-BLG*	12	37 X 28	Copper	Nickel	.089	.115	.131
M16878/27-BMK*	10	105 X 30	Copper	Nickel	.120	.146	.162
M16878/27-BPL*	6	133 X 27	Copper	Nickel	.213	.253	.273

See footnotes at top of next page.

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1/ Inactive for new design.
2/ H.S.C.A. stands for high-strength copper alloy.
3/ C.C. stands for copper-clad.
4/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 5.0 kV.
Impulse dielectric test: 8.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 50,000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 65 ± 1°C (see table II).

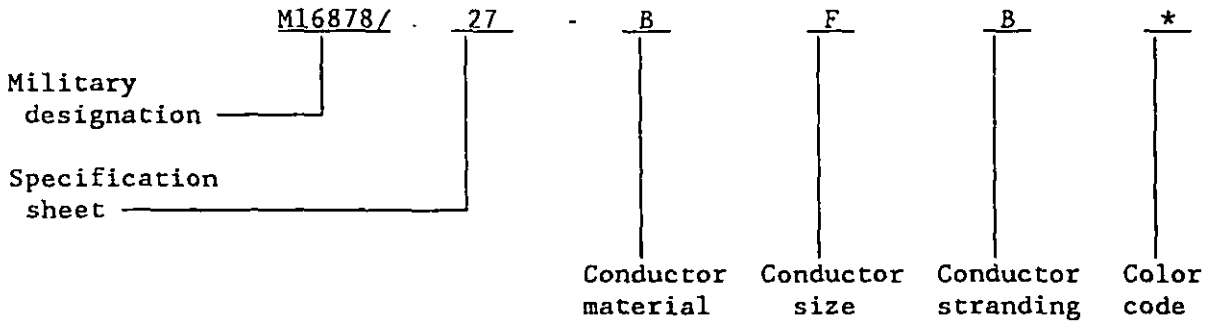
TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter in inches)
32 through 16	1
14, 12	2
10, 8, 6	3

Concentricity: 70 percent (minimum).
Surface resistance: Not Required.
Wrap back: Required.
Heat resistance: Condition at 290°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Shrinkage: Required.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Heat aging: Not required.
Insulation tensile strength: 4000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Not required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Wire length requirements: See appendix B.
Identification of product: Required for wire sizes 22 AWG and larger.

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PIN example (see MIL-W-16878):



SUPERSESSION DATA:

The wire of this specification sheet, by part number, has been replaced and superseded by the wire of MIL-W-22759 as specified in table III.

TABLE III. Supersession by part number.

Part number (former) MIL-W-16878	Part number (replacement) MIL-W-22759
MIL-W-16878/27-BCB	MIL-W-22759/10-2
MIL-W-16878/27-DCB	MIL-W-22759/21-2
MIL-W-16878/27-BDE	MIL-W-22759/10-2
MIL-W-16878/27-DDE	MIL-W-22759/21-2
MIL-W-16878/27-BEE	MIL-W-22759/10-2
MIL-W-16878/27-DEE	MIL-W-22759/21-2
MIL-W-16878/27-BFE	MIL-W-22759/10-2
MIL-W-16878/27-DFE	MIL-W-22759/21-2
MIL-W-16878/27-BGE	MIL-W-22759/10-2
MIL-W-16878/27-DGE	MIL-W-22759/21-2
MIL-W-16878/27-BHE	MIL-W-22759/10-1
MIL-W-16878/27-BJE	MIL-W-22759/10-1
MIL-W-16878/27-BKE	MIL-W-22759/10-1
MIL-W-16878/27-BLE	MIL-W-22759/10-1
MIL-W-16878/27-BMG	MIL-W-22759/10-1
MIL-W-16878/27-BNL	MIL-W-22759/10-8

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-26)

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